Population Memory Rather Than Algorithm Identity Explains Smooth Full Pose IK Tracking in Metaheuristic Ablations

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***Abstract*—Trajectory-level inverse kinematics is a sequence of correlated full-pose optimization problems, so population-based metaheuristics can reuse information across target poses. This simulation-only study asks whether the tracking benefit comes from the optimizer itself (algorithm identity) or from population memory. For a 6-DoF manipulator, we ablate Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) under cold restart, warm-start injection, and, for PSO, full persistence, on a line and a corner trajectory. Warm-start injection reduces roughness, the summed absolute joint change along the trajectory, by a factor of about 19 to 57. With matched injection, PSO and GA are statistically equivalent in roughness and a second-difference proxy, but not demonstrably in accuracy. A bounded trust-region least-squares baseline shows the same memory effect in a classical solver (roughly 95 to 140 times lower roughness). Full PSO persistence adds only a marginal roughness reduction while lowering tight-threshold coverage and corner position accuracy. For this arm and these trajectories, lightweight memory injection, not algorithm identity, is the main mechanism behind smoother simulated tracking; hardware validation remains future work.**

***Index Terms*—inverse kinematics, particle swarm optimization, genetic algorithms, warm start, trajectory tracking, robotics**

I. Introduction

Inverse kinematics (IK) for serial manipulators is nonlinear, multi-solution, and sensitive to singularities and branch switching. Gradient-free metaheuristics such as particle swarm optimization (PSO) and genetic algorithms (GA) are therefore attractive when the objective is nonconvex or constrained by joint limits. In trajectory tracking, however, consecutive targets are correlated, so information from the previous target can be reused.

This creates a methodological risk: if one method preserves its population state while another restarts from scratch, a comparison may credit the nominal algorithm with an effect of memory. This study separates the two factors. *Algorithm identity* denotes the optimizer's own search rules (PSO's velocity update versus GA's selection, crossover, and mutation), whereas *population memory* is the state carried from one target to the next. We cross both factors in full-pose IK tracking of a line and a corner trajectory and measure joint continuity, pose accuracy, and computational effort.

Gradient-based numerical IK, including Levenberg–Marquardt-style solvers, reaches high precision when multi-started [1]. Under the single-start protocol used here as a reference, however, a bounded trust-region solver is local-minimum-limited: its cold starts reach the pose-error threshold of 10^{-3} on only 10–43% of the line targets per run, and its warm starts lock into the first basin found, which motivates global, gradient-free search under joint limits. The contribution is therefore an ablation of a memory mechanism whose effect also appears in a classical solver, not a claim that metaheuristics outperform multi-start numerical IK.

The central result is that warm-start injection, a lightweight memory mechanism, captures essentially the whole improvement in roughness, the summed absolute joint change between consecutive targets (Section III-E). On the line, injection reduces PSO roughness from 242.33 ± 39.22 to 4.22 ± 2.47 rad and GA roughness from 122.74 ± 30.64 to 4.93 ± 3.67 rad ($d = 8.569$, $p_{\mathrm{adj}} = 2.74 \times 10^{-24}$ for PSO). Once both optimizers receive the same injection memory, their roughness is statistically equivalent by two one-sided tests (TOST), whereas their accuracy is not; under the adopted protocol, the main executability effect is therefore memory rather than algorithm identity.

The experiment evaluates mechanisms rather than proposing a new optimizer: it shows what memory buys, what it does not buy, and when it can hurt. Full PSO persistence achieves only marginally lower roughness than injection, at the cost of larger accuracy variance, lower tight-threshold coverage, and worse corner accuracy. All results are simulation-only: roughness and joint finite differences are kinematic proxies for executability, not evidence of reduced actuator stress or torque.

II. Background and Related Work

For a particle i at iteration k, PSO updates velocity and position as

$$v_i^{k+1} = wv_i^k + c_1 r_1 (p_i^k - x_i^k) + c_2 r_2 (g^k - x_i^k), \quad (1)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1}, \quad (2)$$

where w controls inertia, c_1 and c_2 weight cognitive and social attraction, p_i is the particle best, g is the swarm best, and r_1, r_2 are random samples. These state variables make PSO naturally compatible with moving optima, a topic studied in dynamic optimization and adaptive landscapes [2]–[4].

GA instead evolves a population through selection, crossover, mutation, renewal, and elitism. In the implementation evaluated here, roulette selection promotes fitter individuals, arithmetic crossover recombines parent joint vectors, Gaussian mutation perturbs genes, renewal preserves diversity, and elitism carries the best individuals between generations. GA has a population but no velocity or personal-best state, which matters for the experimental design in Section III.

Metaheuristic IK has been studied with GA, PSO, and hybrids [5]–[10]. Full-pose formulations are the harder regime, because position-only IK can hide orientation infeasibility and branch choices [11], [12]. IK reviews and theses also emphasize redundancy, singularity handling, and the multiplicity of valid joint branches [13], [14]; in trajectory tracking, branch switching can produce jagged joint profiles even when Cartesian errors are acceptable, which motivates an explicit continuity analysis.

III. Methodology

This two-factor ablation crosses algorithm identity (PSO versus GA) with population memory (cold restart and injection, plus persistence for PSO), on two full-pose trajectories (line and corner). Each of the five metaheuristic cells has 30 independent runs under one fixed budget; the single-start numerical IK reference has 10 runs.

A. Manipulator and Full-Pose IK

The experiments use our 6-DoF robotic arm with a standard Denavit–Hartenberg model and bounded joint limits. Each target is a full pose (p^d, R^d), not only a Cartesian position. For a candidate joint vector q, position error is $e_{\text{pos}} = \|p^d - p(q)\|_2$ and orientation error is the geodesic angle on $SO(3)$,

$$e_{\text{ori}} = \arccos\left(\text{clip}\left(\frac{\text{tr}((R^d)^T R(q)) - 1}{2}, -1, 1\right)\right). \quad (3)$$

TABLE I
Standard DH parameters and joint limits of our 6-DoF robotic arm.

Joint	a_i (mm)	α_i (rad)	d_i (mm)	$q_{\min}$	$q_{\max}$
1	0	$+\pi/2$	184	$-180°$	$180°$
2	222	0	0	$-110°$	$155°$
3	205	0	0	$-80°$	$210°$
4	0	$-\pi/2$	0	$-180°$	$180°$
5	0	$+\pi/2$	0	$-50°$	$230°$
6	0	0	75	$-180°$	$180°$

The clipping avoids numerical domain errors near identical rotations.

Using the standard DH convention with $\theta_i = q_i$ and all joint offsets zero (Table I), the derived scales are $L_{\text{reach}} = \sum_i \sqrt{a_i^2 + d_i^2} = 686$ mm and $\|q_{\max} - q_{\min}\|_2 = 13.76$ rad, and the initial posture is $q = 0$ for all joints.

B. Dimensionless Fitness

Position error in millimeters and orientation error in radians are not directly commensurate. The objective therefore normalizes each term by a model-derived scale:

$$f(q) = w_p \frac{e_{\text{pos}}}{L_{\text{reach}}} + w_o \frac{e_{\text{ori}}}{\pi} + w_j \frac{\|q - q_{\text{ref}}\|_2}{\|q_{\max} - q_{\min}\|_2}. \quad (4)$$

Here $L_{\text{reach}} = 686$ mm is computed from the DH model at runtime, and the joint-continuity term is normalized by the joint-range norm. We use $w_p = w_o = 1.0$ and $w_j = 0.1$. This adimensionalization is the basis for the weight choice: the two accuracy terms become mutually comparable, while the continuity term carries a smaller, but not negligible, weight. The reference vector q_{ref} is defined identically in every cell: at the first trajectory point it is the initial posture $q = 0$ from Section III-A, and at each later point it is the joint solution executed at the previous point (the incumbent after 160 iterations). Injection seeds its individuals around this same vector. The term is therefore present under cold restart as well, where PSO-cold roughness still reaches 242.33 rad on the line, so the continuity term alone does not produce smooth tracking. A control run with $w_j = 0$ (PSO-cold and PSO-injection, line, 30 runs) kept the memory effect intact (median roughness 256.3 versus 3.42 rad, Cliff's $\delta = 1.0$) and changed no accuracy or continuity metric at $p < 0.05$. The objective is deterministic.

C. Experimental Factors

The ablation contains five cells: PSO-cold, PSO-injection, PSO-persistence, GA-cold, and GA-injection. A cold restart samples a fresh uniform-random population at every trajectory point. Injection also samples a fresh population, but seeds $\lceil 0.25 \cdot 40 \rceil = 10$ individuals around the previous point's best solution using zero-mean Gaussian jitter with standard deviation $\sigma = 0.1$ rad per joint, clamped to the joint limits; at the first point it degenerates to cold start. The injection fraction (one quarter of the population) was a heuristic choice, and $\sigma = 0.1$ rad reuses the scale of the GA mutation; both

were kept fixed across all cells and runs, without tuning or a pilot sweep, and their sensitivity is not evaluated here. Persistence carries a single PSO swarm through the trajectory, including positions, velocities, personal bests, and global best, with memories re-evaluated when the target changes.

PSO used a population of 40 for 160 iterations, fixed inertia $w = 0.7$, $c_1 = c_2 = 1.5$, a global-best topology as in Eqs. (1)–(2), synchronous updates, velocity clamping at $\pm 20\%$ of each joint range, initial velocity sampled from $\mathcal{U}(-0.05, 0.05)$ times each joint range, and position clamping at the joint limits without zeroing or reflecting velocity. GA used a population of 40 for 160 generations, two elites, roulette selection with replacement using probability proportional to $1/(f + 10^{-12})$ (40 draws, paired in sequence), arithmetic crossover with probability 0.7 per pair ($\alpha \sim \mathcal{U}(0, 1)$ drawn per pair, producing children $\alpha p_1 + (1 - \alpha)p_2$ and $(1 - \alpha)p_1 + \alpha p_2$), per-gene Gaussian mutation with probability 0.2 and standard deviation $\sigma = 0.1$ rad followed by clamping, and replacement of the worst $\lfloor 0.2 \cdot 38 \rfloor = 7$ non-elite individuals by uniform-random individuals at every generation.

Persistence is PSO-only by design. It denotes carrying velocity and personal-best state, for which GA has no principled direct equivalent, and an artificial GA persistence variant would blur the factor under study; the design is therefore intentionally unbalanced and analyzed with planned comparisons rather than a full-crossed ANOVA. All experiments use a static fitness landscape; the adaptive weight schedule supported by the framework is left for future work.

D. Trajectories and Reachability Guard

Both trajectories contain 30 full-pose targets spaced uniformly in Cartesian arc length. The line (250 mm, 8.62 mm per step) preserves comparability with preliminary experiments by the authors. The corner (350 mm, 11.5–12.5 mm per step) has a sharp XY corner at its midpoint and is deliberately adversarial for memory, because a warm start may delay recovery after the direction changes. Orientation rotates from $+15^\circ$ to -15° about z by quaternion SLERP over the normalized arc length (1.03° per step); no time parameterization is imposed. Every target was certified reachable beforehand by bounded multi-start numerical IK to a dimensionless tolerance of 10^{-3}, so reported tracking errors are not pose infeasibility.

Because roughness and the second-difference proxy sum per-step differences, their absolute values depend on the number and spacing of trajectory points; per step, PSO roughness falls from 8.36 rad/step (cold) to 0.15 rad/step (injection) on the line. The injection jitter ($\sigma = 0.1$ rad) is about four times the mean per-joint displacement between warm-started solutions (about 0.025 rad), so coarser sampling could weaken the memory benefit; this was not tested.

E. Budget, Effort, Metrics, and Statistics

Each point runs to a fixed cap of 160 iterations with population size 40, because early stopping would let a method seed a different next-point trajectory and contaminate the continuity metrics. The budget is matched in iterations, not in objective evaluations: PSO uses 6,440 evaluations per point in every memory mode, while GA uses 12,520, of which 7,240 are distinct candidates, because its renewal step scores non-elites and then re-evaluates the whole population. Effort-to-reach-ε is extracted post hoc from recorded incumbent histories. The threshold uses pose error only, excluding the continuity term, because including a term that rewards moving less would re-confound memory with accuracy. A point is counted as converged when its pose error $e_{\mathrm{pos}}/L_{\mathrm{reach}} + e_{\mathrm{ori}}/\pi$ falls below ε; coverage is the fraction of individual points that reach ε. Runs not reaching ε are right-censored at 160 and flagged; coverage fractions make censoring visible. When two cells are fully censored, samples are constant and the corresponding t-test is undefined, so the result is interpreted as neither cell reaching the threshold and coverage carries the information.

For the executed joint path $q_k \in \mathbb{R}^6$, $k = 1, \ldots, 30$, roughness is the summed absolute joint change over consecutive targets,

$$R = \sum_{k=1}^{29} \sum_{j=1}^{6} |q_{j,k+1} - q_{j,k}|, \tag{5}$$

and the second-difference proxy is

$$A = \sum_{k=1}^{28} \sum_{j=1}^{6} |q_{j,k+2} - 2q_{j,k+1} + q_{j,k}|. \tag{6}$$

Accuracy is the RMSE of position and orientation over the 30 targets; the peak absolute first and second differences are also reported. All finite differences use a unit index increment rather than a time interval, so they measure spatial joint-space continuity, not velocity or acceleration. Each cell has 30 independent runs (seeds 42–71), all from the final study; no pilot runs are included. Because the design lacks GA-persistence, inference uses planned comparisons aligned with the research questions: Q1, memory within PSO, and Q2, algorithm under matched memory. Welch tests and Cohen's d are used for the uncensored roughness, second-difference, and wall-time families; Mann–Whitney U, Cliff's δ, and Kruskal–Wallis omnibus tests are used for heavy-tailed accuracy and effort families. Benjamini–Hochberg FDR correction is applied per metric family. Because a core claim is equivalence, non-significance is not treated as sufficient: TOST uses bounds anchored to the cold-to-injection memory effect, rather than an independent practical tolerance. We therefore also report the 90% CI of the difference and the smallest bound at which equivalence holds. BCa bootstrap confidence intervals use 10000 resamples.

The bounded trust-region reflective (TRF) least-squares baseline uses the same solver as the reachability guard, SciPy's `least_squares` (SciPy 1.18, Python 3.14) with method `trf` and joint limits [15], under a matched single-start protocol. TRF-cold uses a random initial guess at every point; TRF-warm uses the previous solution as the next initial guess. Its ten runs (seeds 42–51) are reported descriptively, without inferential tests against the metaheuristic cells, because of the different repetition count and near-deterministic within-basin behavior.

IV. Results

A. Accuracy

Table II summarizes accuracy, continuity, and runtime. Because accuracy is heavy-tailed, the analysis relies on medians and rank-based tests, with Welch/Cohen summaries as secondary checks.

For position accuracy, medians reverse the mean-only story. On the line, PSO-injection has median position RMSE 0.60 mm (IQR 31.5), compared with PSO-cold median 61.0 mm (IQR 13.8); Mann–Whitney testing gives $p_{\text{adj}} = 1.0 \times 10^{-3}$ and Cliff's $\delta = 0.536$ (secondary Welch summary: $d = 0.833$). About half of the line PSO-injection runs end below 1 mm, while 14 of 30 have position RMSE above 10 mm, so the median (0.60 mm) lies at the boundary of a two-mode distribution and should not be read as a typical error. With memory, failures typically appear as a single contiguous block of failed points lasting to the end of the trajectory (0.5–1.5 episodes per run, versus 3.7–7.5 under cold restart), the same lock-in behavior seen in the warm single-start baseline. Runs ending with a failed point are 12/30 for PSO-injection, 14/30 for PSO-persistence, and 7/30 for GA-injection on the line, and 12/30, 20/30, and 8/30, respectively, on the corner. Under matched injection on the line, PSO and GA position RMSE medians are 0.60 mm (IQR 31.5) and 1.85 mm (IQR 19.5), with no rank-significant difference ($p_{\text{adj}} = 0.17$, $\delta = -0.218$); this is not an equivalence result. On the corner, the persistence penalty is rank-supported: PSO-injection versus PSO-persistence medians are 5.41 mm (IQR 91.3) and 19.8 mm (IQR 161), with $p_{\text{adj}} = 5.1 \times 10^{-3}$ and $\delta = -0.458$. Under cold start, GA remains more accurate than PSO on the corner by ranks, with medians 33.2 mm (IQR 17.5) versus 66.6 mm (IQR 15.3), $p_{\text{adj}} = 1.6 \times 10^{-9}$ and $\delta = 0.976$ (secondary Welch summary: $d = 2.990$). Kruskal–Wallis omnibus tests over the PSO memory triple are significant for every accuracy metric on both trajectories ($p \leq 1.6 \times 10^{-3}$).

For orientation accuracy, there is no uniform winner. Line ranks favor PSO-injection over GA-injection ($p_{\text{adj}} = 0.016$, $\delta = -0.39$), whereas the corner bootstrap mean difference for PSO-injection minus GA-injection is 0.0072 rad with 95% BCa CI $[0.0002, 0.0243]$, favoring GA-injection. Memory remains useful within PSO on the line: PSO-cold versus PSO-persistence gives a secondary mean-based summary of $d = 1.373$ and $p_{\text{adj}} = 6.74 \times 10^{-5}$.

B. Kinematic Executability Proxies

Injection removes most of the roughness in both algorithms (Table II); the PSO cold-versus-injection effect on the line is $d = 8.569$ with $p_{\text{adj}} = 2.74 \times 10^{-24}$, and the corresponding second-difference effect is $d = 8.541$ with the same adjusted p value. Fig. 1(a)–(b) shows the mechanism: cold restarts switch joint branches between consecutive targets, whereas injection keeps each joint on a smooth path.

The keystone comparison is algorithm under matched memory. GA-cold has lower roughness than PSO-cold on the line, although both cold-start conditions remain rough ($d = 3.398$, $p_{\text{adj}} = 2.76 \times 10^{-18}$). Once injection is applied to both algorithms, TOST establishes equivalence for roughness and the second-difference proxy on both trajectories, rather than merely failing to detect a difference. The $\pm 10\%$ bounds, anchored to the smaller memory effect, are ± 11.78 rad (line) and ± 9.56 rad (corner) for roughness and ± 22.02 and ± 17.91 rad/step^2 for the second-difference proxy; all TOST p-values are below 10^{-17}, and the bootstrap CIs for Cohen's d lie within $[-0.668, 0.315]$ for roughness and $[-0.651, 0.470]$ for the second-difference proxy. Equivalence also holds at much tighter margins: 2.06 rad (line) and 1.75 rad (corner) for roughness, and 3.65 and 2.66 rad/step^2 for the second-difference proxy. These margins are 1.49–1.83% of the smaller memory effects and below the run-to-run standard deviations within each cell, so differences of that size cannot be distinguished from replication noise. Accuracy is different: position and orientation RMSE are not equivalent under the same analysis, and except for the corner orientation bootstrap CI, neither a difference nor equivalence is demonstrated. Persistence adds a marginal roughness reduction relative to injection on both line ($d = 0.693$, $p_{\text{adj}} = 1.57 \times 10^{-2}$) and corner ($d = 0.913$, $p_{\text{adj}} = 1.30 \times 10^{-3}$), small compared with the cold-to-injection change. A semicircular arc run under the same protocol (not tabulated) repeats the continuity pattern: injection lowers roughness 37.8 times for PSO and 14.5 times for GA, and the two remain equivalent (TOST at ± 8.17 rad), although 73% of PSO-injection arc runs end above 10 mm RMSE.

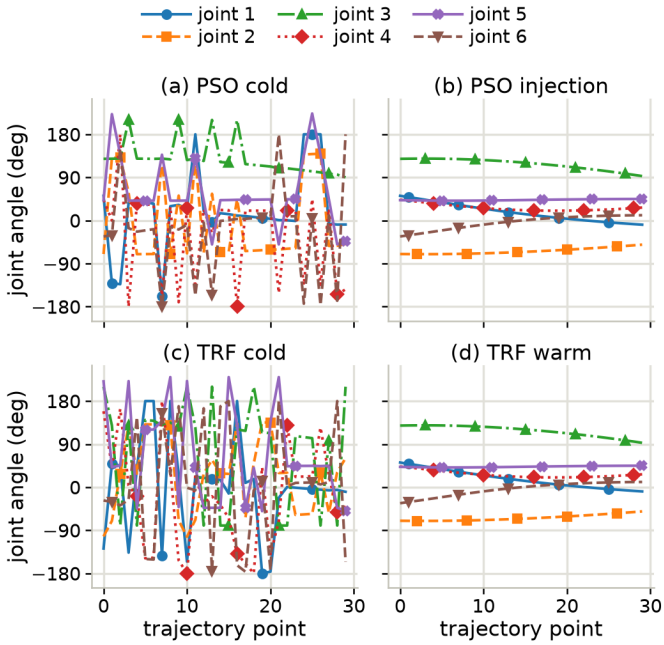


Fig. 1. Lowest position-RMSE run per condition on the line (30 seeds for PSO, 10 for TRF): best cases, not typical runs. Cold restarts switch branches; injection and warm starts do not.

C. Effort and Coverage

Table III reports iterations to reach pose-error thresholds. Effort clarifies what memory does not buy: it improves

TABLE II

ACCURACY, CONTINUITY, AND RUNTIME FOR THE METAHEURISTIC CELLS (30 RUNS) AND THE SINGLE-START TRF BASELINE (10 RUNS). MEAN ROWS REPORT MEAN ± SD; MEDIAN ROWS REPORT MEDIAN (IQR). A FAILED RUN HAS POSITION RMSE ABOVE 10 MM; A FAILED POINT HAS POSITION ERROR ABOVE 10 MM. N/A: NOT REPORTED FOR THE BASELINE.

Traj.	Metric	PSO cold	PSO inj.	PSO pers.	GA cold	GA inj.	TRF cold	TRF warm
Line	Position RMSE (mm)	61.66 ± 12.83	31.74 ± 49.12	33.27 ± 44.59	47.56 ± 13.27	25.04 ± 42.38	100.9 ± 10.7	93.3 ± 70.0
	Position RMSE med. (mm)	61.04(13.8)	0.60(31.5)	7.80(62.2)	48.19(17.6)	1.85(19.5)	102.6(10.8)	101.2(39.6)
	Runs with position RMSE > 10 mm (%)	100	46.7	46.7	100	36.7	n/a	n/a
	Points with pose error $< 10^{-3}$ (%)	51.8	63.9	37.3	3.2	20.1	28.7	18.7
	Orientation RMSE (rad)	0.0546 ± 0.0509	0.0105 ± 0.0197	0.0045 ± 0.0086	0.0562 ± 0.0496	0.0066 ± 0.0106	0.348 ± 0.097	0.179 ± 0.135
	Roughness (rad)	242.33 ± 39.22	4.22 ± 2.47	2.91 ± 1.03	122.74 ± 30.64	4.93 ± 3.67	328.5 ± 39.7	3.47 ± 2.70
	Second-difference proxy (rad/step^2)	432.53 ± 70.97	2.73 ± 5.28	1.59 ± 0.72	223.99 ± 55.12	3.82 ± 6.56	572.2 ± 71.7	1.81 ± 5.22
	Peak joint step (rad/step)	6.00 ± 0.49	0.306 ± 0.765	0.104 ± 0.068	4.71 ± 0.99	0.379 ± 0.729	6.08 ± 0.379	0.401 ± 1.12
	Peak second difference (rad/step^2)	10.00 ± 1.35	0.287 ± 0.774	0.105 ± 0.074	8.00 ± 1.07	0.346 ± 0.721	11.23 ± 1.16	0.374 ± 1.14
	Wall time (s)	10.23 ± 0.17	10.56 ± 0.15	14.96 ± 0.65	30.99 ± 2.17	25.06 ± 2.45	0.394 ± 0.051	0.183 ± 0.058
Corner	Position RMSE (mm)	67.30 ± 10.45	46.62 ± 69.32	76.17 ± 90.15	33.22 ± 12.26	27.75 ± 53.93	112.2 ± 14.9	85.4 ± 66.6
	Position RMSE med. (mm)	66.63(15.3)	5.41(91.3)	19.8(161)	33.2(17.5)	2.80(10.1)	110.6(10.7)	85.7(129)
	Runs with position RMSE > 10 mm (%)	100	50.0	73.3	96.7	40.0	n/a	n/a
	Points with pose error $< 10^{-3}$ (%)	59.4	63.0	21.3	3.4	26.9	27.0	29.0
	Orientation RMSE (rad)	0.0852 ± 0.0394	0.0123 ± 0.0291	0.0262 ± 0.0505	0.0586 ± 0.0475	0.0051 ± 0.0055	0.420 ± 0.106	0.383 ± 0.576
	Roughness (rad)	223.81 ± 39.26	4.61 ± 2.39	2.76 ± 1.60	100.80 ± 39.60	5.20 ± 2.95	357.1 ± 26.2	2.52 ± 1.10
	Second-difference proxy (rad/step^2)	401.12 ± 71.55	3.26 ± 4.96	1.61 ± 1.25	182.69 ± 74.75	3.56 ± 5.95	618.1 ± 45.4	0.357 ± 0.197
	Peak joint step (rad/step)	5.72 ± 0.66	0.357 ± 0.811	0.105 ± 0.076	4.17 ± 1.03	0.377 ± 0.770	6.22 ± 0.128	0.059 ± 0.014
	Peak second difference (rad/step^2)	9.76 ± 1.59	0.385 ± 0.809	0.104 ± 0.083	7.23 ± 1.18	0.396 ± 0.779	11.48 ± 1.00	0.060 ± 0.036
	Wall time (s)	10.49 ± 0.09	10.50 ± 0.12	10.80 ± 0.17	26.41 ± 3.82	29.09 ± 2.23	0.377 ± 0.035	0.252 ± 0.121

continuity and orientation accuracy, but not uniformly faster convergence to tight positional tolerances. On the line at $\varepsilon = 10^{-3}$, PSO persistence requires 147.40 ± 19.65 iterations, versus 120.08 ± 37.34 for PSO cold and 100.18 ± 49.69 for PSO injection (cold versus persistence: Cliff's $\delta = -0.336$, $p_{\mathrm{adj}} = 2.27 \times 10^{-2}$). On the corner at the same threshold, persistence is nearly fully censored, requiring 157.13 ± 9.21 iterations versus 91.87 ± 25.71 for cold ($\delta = -0.879$, $p_{\mathrm{adj}} = 3.60 \times 10^{-9}$). At the moderate $\varepsilon = 10^{-2}$ threshold on the line, GA-injection reaches accuracy faster by ranks than PSO-injection, with medians 17.75 versus 29.25 iterations ($p_{\mathrm{adj}} = 9.4 \times 10^{-5}$, $\delta = 0.624$).

Coverage at tight thresholds strengthens this interpretation. On the line at $\varepsilon = 10^{-4}$, PSO-injection reaches about 0.64 of trajectory points, PSO-cold about 0.51, and PSO-persistence about 0.30; GA cells reach none. Thus persistence can settle into a basin that is smooth but less able to refine to a tight pose threshold (Fig. 2). GA coverage at $\varepsilon \leq 10^{-3}$ is also budget-limited. Among censored points at $\varepsilon = 10^{-3}$, 84.3% (line) and 87.8% (corner) of GA-cold points and 48.4% and 58.2% of GA-injection points still improved by more than 10% between iterations 120 and 160. For PSO-cold and PSO-injection this share is at most 2.1%, and for PSO-persistence 17.9% (line) and 12.1% (corner). A larger budget was not tested.

D. Computational Cost

Cost is reported as function evaluations per trajectory point, a machine-independent measure. PSO uses 6,440 evaluations in every memory mode (40 initial evaluations, or 40 re-evaluations of the stored personal bests under persistence, plus 160 iterations of 40), so injection and persistence add no evaluations relative to cold restart. GA uses 12,520 (40 plus 160 generations of 78), 1.94 times as many: each generation scores the 38 non-elites to select replacements and then re-evaluates all 40 individuals. Because fitness is deterministic, 33 of these 78 calls repeat points already evaluated, so GA evaluates 7,240 distinct points per trajectory point, 1.12 times

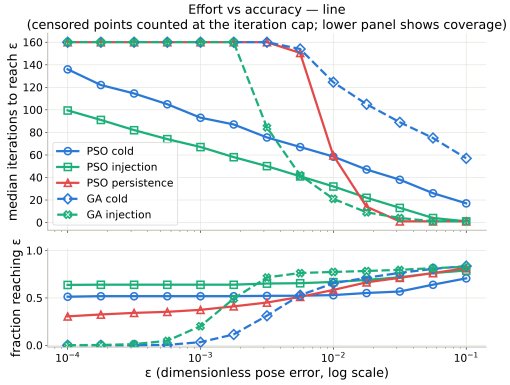


Fig. 2. Line effort-versus-ε and coverage. Persistence has low roughness but worse tight-threshold coverage, so memory is not universally faster convergence.

PSO. GA's wall time is 2.4–3 times PSO's (on the line, 30.99 ± 2.17 s for GA-cold versus 10.23 ± 0.17 s for PSO-cold), consistent with per-call costs of about 67–83 μs for GA and 53–55 μs for PSO; wall-time differences between cells with identical evaluation counts are not interpreted as algorithmic costs.

V. DISCUSSION

The ablation indicates that, under the adopted iteration budget, injection accounts for most of the continuity gain: line roughness falls by about 96–98%, and matched-injection PSO and GA satisfy the reported TOST bounds for roughness and second differences. This supports a memory effect for these kinematic proxies, not equivalence in accuracy or a universal ranking of optimizers.

Position accuracy remains unstable relative to published sub-millimeter results [11], [12]: the best mean RMSE here is 25.04 mm (3.6% of reach), which reflects runs locking into a poor basin (Section IV-A). All cells use 160 iterations, population 40, no per-cell tuning, and a continuity term that

TABLE III

ITERATIONS TO REACH POSE-ERROR THRESHOLD ε (CAP 160). PER RUN, EFFORT IS THE MEDIAN OVER THE 30 POINTS; MEAN ROWS REPORT MEAN $\pm$ SD AND MEDIAN ROWS MEDIAN (IQR) ACROSS RUNS.

ε	Traj.	PSO cold	PSO inj.	PSO pers.	GA cold	GA inj.
10^{-3}	Line	120.08 ± 37.34	100.18 ± 49.69	147.40 ± 19.65	160.0 ± 0.0	160.0 ± 0.0
10^{-3} med.	Line	115.25(76.4)	62.00(100.0)	160.00(13.1)	160.00(0.0)	160.00(0.0)
10^{-2}	Line	100.35 ± 53.94	72.78 ± 62.93	74.52 ± 62.10	125.10 ± 13.49	46.58 ± 57.93
10^{-2} med.	Line	72.50(109.9)	29.25(132.9)	34.50(132.6)	123.25(16.0)	17.75(8.4)
10^{-3}	Corner	91.87 ± 25.71	90.80 ± 46.20	157.13 ± 9.21	160.0 ± 0.0	159.97 ± 0.18
10^{-3} med.	Corner	80.75(10.9)	61.25(100.9)	160.00(0.0)	160.00(0.0)	160.00(0.0)
10^{-2}	Corner	63.58 ± 36.03	67.98 ± 61.31	120.37 ± 52.30	120.27 ± 12.05	37.28 ± 49.01
10^{-2} med.	Corner	48.25(6.1)	29.00(132.0)	160.00(100.1)	119.25(15.0)	18.50(4.9)

can trade position accuracy for smoothness; the protocol compares mechanisms rather than maximizing accuracy.

Persistence lowers roughness only marginally beyond injection, while reducing tight-threshold coverage and increasing accuracy variance. On the adversarial corner it also worsens median position RMSE, ends 20/30 runs in a failed point, and is nearly fully censored at $\varepsilon = 10^{-3}$: the same memory that limits branch switching can delay recovery after a sharp direction change.

Algorithm comparisons must therefore remain memory-matched and budget-qualified. Under cold start, GA is more accurate than PSO on both trajectories by ranks ($p_{\mathrm{adj}} \leq 5.7 \times 10^{-4}$), but it uses 1.94 times as many objective evaluations (1.12 times as many distinct candidates) and runs 2.4–3 times slower. Under matched injection, the line position-RMSE difference is not significant; these results do not identify a universally better optimizer.

The TRF baseline shows that warm-start continuity is not specific to population search: TRF-warm lowers roughness by roughly 95–140 times on both trajectories, and the corner peak joint step falls from 6.22 to 0.059 rad/step. Yet single-start TRF remains basin-dependent, with TRF-cold position RMSE above 100 mm on both trajectories. Although TRF is roughly 25–60 times faster than PSO, the time for one PSO target would allow about 27 TRF starts; a budget-matched multi-start TRF baseline was not evaluated.

VI. CONCLUSION AND FUTURE WORK

This simulation-only study finds that warm-start injection captures the dominant continuity benefit in full-pose IK for the tested 6-DoF arm. With injection, PSO and GA are equivalent in roughness and the second-difference proxy by TOST, while accuracy equivalence is not established and GA uses 1.94 times as many function evaluations. Full PSO persistence offers only marginally lower roughness and costs tight-threshold coverage, accuracy stability, and corner position accuracy.

These conclusions are limited to one manipulator, two tabulated 30-point trajectories, fixed optimizer parameters, and a 160-iteration budget. Future work should test other robots, other curved paths, poses near joint limits or singularities, other discretizations and budgets, budget-matched multi-start numerical IK, and the injection parameters, and should validate the kinematic proxies on hardware before claims about physical execution.

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